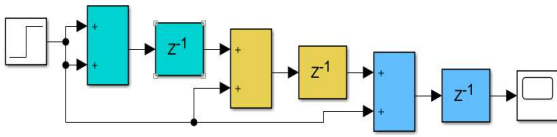


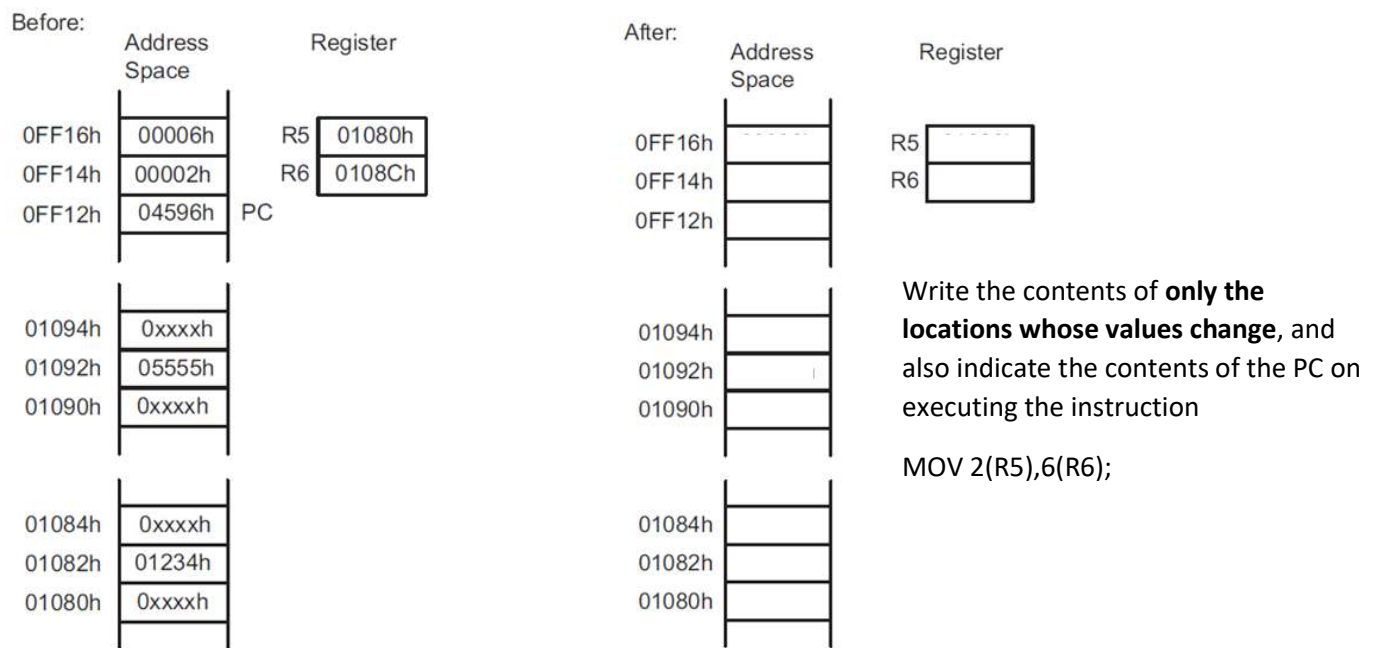
Sample questions – short answer. MCQ

- 1) Plot the output of the system of adders and their associated delays shown below if each (z^{-1}) delay is 10 ns and input changes from 0 to 1 after system is stable ($t=0$)



- 2) The _____ points to the next instruction to be fetched
- 3) _____ resets the processor if the program becomes stuck in an infinite loop
- 4) Volatile memory is used to store data whose value must be retained when power is removed (T/F) _____
- 5) The number of address lines required to address 4MB of memory is _____
- 6) It can be programmed electrically but not erased. Devices must be exposed to ultraviolet (UV) light for about ten minutes to erase them. _____
- 7) A higher voltage is needed to write to flash memory than is necessary for normal operation (T/F) _____
- 8) In _____ memory architecture, the data and program memories are treated as separate systems, each with its own address and data bus
- 9) The contents of memory are shown in the debugger, and machine code is shown in the _____ view
- 10) _____ language: Little more than machine code translated into English mnemonics
- 11) _____ translates the mnemonics into machine code.
- 12) _____ translates high-level language into machine code that the CPU can process.
- 13) In describing the instruction set, RISC stands for _____
- 14) In the MSP 430, the main address bus is _____ bits wide, and the main data bus is _____ bits wide
- 15) Runs on development processor, but executes instructions of target processor _____
- 16) Current drawn by the MSP 430 – Standby Mode: _____ - Off Mode (RAM Retention): _____
- 17) MSP 430 - Only _____ core instructions and _____ addressing modes
- 18) A low-frequency _____ is driven directly from a common 32-kHz watch crystal (write both name and abbreviation)
- 19) DCO stands for _____. The role of the DCO is to _____
- 20) The MSP430 uses a _____ memory architecture
- 21) On which of these memory locations can data access be performed? (Circle) 0x1234 0x1233 0x12AB 0x12AC
- 22) To which of these memory locations can 16 bit peripherals map? (Circle) 0x0234 0x0033 0x00AB 0x00AC
- 23) To which of these memory locations can 8 bit peripherals map? (Circle) 0x0034 0x0033 0x000B 0x000C
- 24) The lowermost memory locations in the MSP map _____
- 25) For 64 kB or lower MSP 430 devices, the highest priority interrupt vector is mapped at the memory location _____
- 26) RAM starts at _____. The end address of RAM depends on the amount of RAM present and varies by device
- 27) If the reset vector's content is _____, the device will be disabled for minimum power consumption

- 28) Ordering where the low-order byte is stored at the lower address and the high-order byte at the higher address_____
- 29) _____instruction set architecture has every instruction usable with every addressing mode
- 30) _____ provides six most used immediate values and reduces code size
- 31) Number of registers in the MSP 430 CPU in all _____
- 32) The _____ is used by the CPU to store the return addresses of subroutine calls and interrupts
- 33) Byte operation: Status bit N is set to the value of bit _____ of the result.
- 34) V bit in status register for SUB(.B), SUBC(.B), CMP(.B) Set when: _____ & _____
- 35) The MSP430 assembler supports, besides the core instructions, 24 additional _____
- 36) The _____ circuit is implemented to provide the proper internal reset signal to the device during power on and power off
- 37) The MSP430 has one active mode and _____ software selectable low-power modes of operation
- 38) There is a low power mode where the CPU and all clocks are disabled and crystal oscillator is stopped (T/F)_____
- 39) The seven addressing modes i) ii)
III) iv) v)
vi) vii)



- 40)
- 41) On executing the instruction P1DIR = 0x7B, which pins are inputs, and which are outputs?
- 42) All jump instructions require _____ code word, and take _____ CPU cycles to execute
- 43) The system reset circuitry shown sources both a _____ and a _____ signal
- 44) There are _____ types of interrupts (* **many** is not a valid answer!)
- 45) The address of the non-maskable interrupt vector is _____.